# POWERFUL TEACHING

## UNLEASH THE SCIENCE OF LEARNING

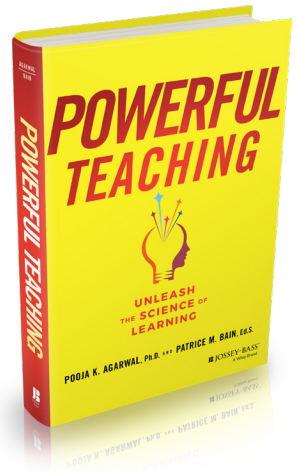
# DO IT YOURSELF RETRIEVAL GUIDE

We practice what we preach. In this Do It Yourself (DIY) Retrieval Guide, we provide the Power Ups featured throughout the book, as well as new discussion questions and exercises for professional development. Keep in mind:

- This is your DIY Retrieval Guide! You can quiz yourself while you read this book, after you read it, or both. You might even want to complete these questions in pencil so you can erase your responses and quiz yourself again down the road for spaced retrieval.
- We intentionally don't provide summaries at the end of our chapters. We don't even give you the answers or page numbers in this DIY Retrieval Guide, so you'll need to retrieve and seek elaborative feedback for yourself!

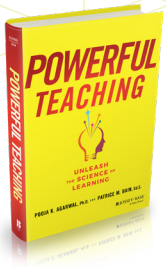
In addition to unleashing the science of learning for yourself, we encourage you to use this DIY Retrieval Guide in a few different ways:

- As a resource for questions, strategies, and implementation checklists in your classroom
- As discussion questions and Power Ups for professional development and book clubs
- As activities and exercises for teachers, administrators, students, and parents
- As a method for modeling Power Tools by experiencing them yourself




Lastly, here are two additional things to note:

- In line with research, we've included a variety of multiple-choice, short answer, conceptual, and higher order questions to boost learning.
- Many of the Power Ups and questions included in this DIY Retrieval Guide were spaced throughout the book; here, we've grouped them by chapter for convenience, but we hope you'll space and interleave the questions for yourself!




# INTRODUCTION TO POWERFUL TEACHING

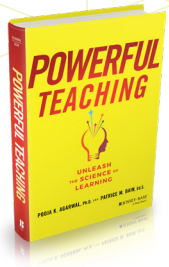
What are five words that come to mind when you think about learning?

1. _____
2. _____
3. _____
4. _____
5. _____

Have you heard of research from the science of learning before? If so, from where? If not, why do you think you haven't?

Have you ever gone gone along with a fad, anecdote, or new idea for your instruction? Why?

What does powerful teaching look like to you?




# CHAPTER 1: DISCOVER THE POWER BEHIND POWER TOOLS

Describe the four Power Tools in your own words.

1. _____

2. _____

3. _____

4. _____

Describe these key terms in your own words:

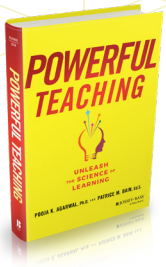
| | |
|---|---|
| desirable difficulties | |
| cognitive science | |
| the science of learning | |

How do these terms relate to education?

What's the difference between **encoding** and **retrieval**?

When we talked about three stages of learning (encoding, storage, retrieval) in the context of going to a coffee shop. Describe the three stages of learning in a situation from your everyday life!

Take a moment and think about how you teach. What do students do in your classroom each day? Now, ask yourself: Are you supporting their short-term learning or long-term learning?




# CHAPTER 2: BUILD A FOUNDATION WITH RETRIEVAL PRACTICE

Why are desirable difficulties a good thing?

Why is having students retrieve information more powerful for learning than having them listed or read something?

Where do you focus your teaching: encoding, storage, or retrieval? Do you focus on one stage more than another stage?

In the Roediger and Karpicke (2006) study we presented, did college students predict they would remember more after re-reading or after retrieval practice? Why?

Research studies we present often include a re-study condition. Why is a re-study condition important? Name at least one reason. If that was easy, challenge yourself: name two reasons!

How do you define retrieval practice? How would you describe retrieval practice to parents?

In this chapter, we listed 10 benefits of retrieval practice. What are three of them?

1. 
2. 
3. 

What's the difference between retrieval practice, formative assessment, and summative assessment?

What is one or more reasons why transfer of knowledge is challenging for students?

What are two examples of retrieval practice in your everyday life where you purposely retrieve in order to remember something (rather than focusing on encoding it)?

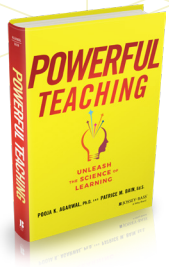
1.

2.

Complete the table below, based on the three stages of learning: encoding, storage, and retrieval.

|  | Did you intentionally try to **encode** this information? | How long have you had this information **stored**? | How many times have you **retrieved** this information? |
|---|---|---|---|
| What you ate for breakfast | | | |
| King Tut's age | | | |
| Your favorite vacation | | | |
| One thing about this book | | | |

Lastly, circle the boxes where you feel your own learning occurred. Do you see any patterns in what you circled? When it comes to the classroom, where do you focus your activities and teaching practices – encoding, storage, or retrieval? Do you focus on one stage more than another stage?




# CHAPTER 3: EMPOWER TEACHING WITH RETRIEVAL PRACTICE STRATEGIES

Why should retrieval practice be used as a learning strategy and not an assessment strategy?

Do a Brain Dump about Brain Dumps! Write down everything you know about Brain Dumps.

Last chapter, we asked you to define retrieval practice in your own words to parents. Now, take a moment and ponder: how would you describe retrieval practice to students?

In the 2006 study by Roediger and Karpicke, mentioned in the previous chapter, how did college students engage in retrieval practice?

As we discussed in the previous chapter, what does it mean to move from pyramid learning to powerful learning? Write a description or draw your own diagram!

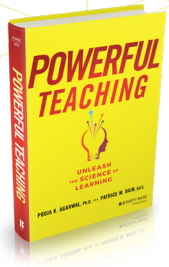
As a powerful teacher, here are questions you might encounter from teachers, administrators, and parents. How would you respond to each one?

- Is retrieval practice effective for only basic information, like vocabulary and facts?
- Should retrieval practice questions be multiple-choice, short answer, or free response?
- Are Pre-Tests before a lesson beneficial for learning?
- Which is more effective: open-book or closed-book quizzes?
- Is retrieval practice effective if students generate their own questions?
- Are active strategies, like concept mapping, as effective as retrieval practice?
- How is retrieval practice different from "cold calling?"
- Don't we have to start with encoding and get information in, before students can retrieve and pull information out?
- Can't I just give cumulative exams? Isn't that the same thing as spacing and interleaving?

Retrieval practice can take three minutes or less! How can you incorporate just one of these strategies and start small tomorrow?

Which of these Power Tool strategies could you incorporate into your teaching easily? Which one works best for you, given your content and your classroom?

**A.** Brain Dumps **B.** Two Things **C.** Mini-Quizzes **D.** Dice Game




# CHAPTER 4: ENERGIZE LEARNING WITH SPACING & INTERLEAVING

Why do you think your students cram the night before an exam?

One additional benefit of spacing is savings. What is it? Give an example or describe it in your own words.

Why is spacing more effective than cramming? Why is interleaving more effective than blocking?

How do you already use spacing in your classroom? How can you incorporate even more?

How much spacing is optimal for learning?

Which type of spaced retrieval practice is most beneficial for long-term learning?

**A.** Retrieval before a lesson

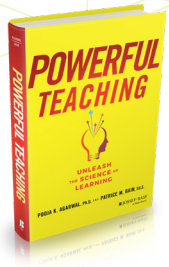
**B.** Retrieval during a lesson

**C.** Retrieval immediately after a lesson

**D.** Retrieval a few days after a lesson

What's the difference between spacing and interleaving?

What is one example in your everyday life where you purposely **retrieve** information in order to remember it, rather than focusing on **encoding** to remember it?




# CHAPTER 5: ENGAGE STUDENTS WITH FEEDBACK-DRIVEN METACOGNITION

What is a judgment of learning? What is a confidence rating?

| judgement of learning | confidence rating |
| --- | --- |
| | |

- First, on a scale from 0 - 100%, what percentage of the previous chapter on spacing and interleaving will you remember in one week?

0% 50% 100%

- Second, who invented the polio vaccine?

- Rate how you feel on a scale from 1 (guessing) to 5 (definitely sure).

(guessing) 1 5 (definitely sure)

- Now, which of these questions is a judgment of learning and which is a confidence rating?

In this chapter, we presented a 2008 study by Agarwal and colleagues where students predicted how much they would remember after one week. Retrieve and write down Two Things you remember about that study!

1.

2.

Take a moment and think about metacognition in your classroom. What does it look like? Now, ask yourself:

- Do my students really know it or do they simply think they do?
- Have my students assessed their own learning through retrieval, or are they relying on their intuition that something was easy to learn, therefore it'll be easy to remember?
- Do my students seem confident in their knowledge, and if so, what may be contributing to this confidence?

In an earlier chapter, we presented a study where students' predictions about how much they would remember after one week mismatched their actual performance after one week. Retrieve and write down Two Things about that study.

1.

2.

What are the Four Steps to Metacognition?

1. _____

2. _____

3. _____

4. _____

What's the difference between Retrieval Cards and Metacognition Sheets?

When you provide students with feedback, are you also giving them opportunities to examine their metacognition? Or are they simply looking at a grade and moving on?

| Who was Lady Murasaki Shikibu? | What is the capital of Australia? | What is the capital of Kentucky? |
|---|---|---|
| | | |

› Do you know the answers or are you guessing? On a scale from 1 (guessing) to 5 (definitely sure), how confident are you in your answers?

**(guessing) 1**  **5 (definitely sure)**

In Figure 12-1, we've provided a few "items to know" about Power Tools. Complete this Metacognition Sheet using the Four Steps of Metacognition. Provide this activity during professional development, too!

**Metacognition Sheet**

| ★ | ? | Items to Know | Answer |
|---|---|---|---|
| | | The difference between spacing and interleaving | |
| | | The illusion of fluency | |
| | | The most effective types of feedback for transfer | |
| | | Why retrieval practice improves learning | |

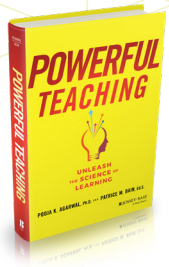
Figure 12-1: Use the Four Steps of Metacognition and explore what you've learned from this book!

When it comes to providing students with feedback, which of these options is true?

**A.** It's optimal to provide students with feedback quickly, even while they are retrieving. This way, they won't make any mistakes.

**B.** It's optimal to provide students with feedback that includes elaborative examples and descriptions.

**C.** It's optimal to provide students with after a delay. Immediate feedback doesn't improve learning; only delayed feedback improves learning.




# CHAPTER 6: COMBINE POWER TOOLS AND HARNESS YOUR TOOLBOX

For each of the four Power Tools, write down how you are already applying it in your instruction or curriculum.

Write down four things:

- One thing teachers, students, or parents should know about the science of learning
- One thing teachers, students, or parents should know about retrieval practice
- One thing teachers, students, or parents should know about metacognition
- One thing teachers, students, or parents should know about a research study presented in this book

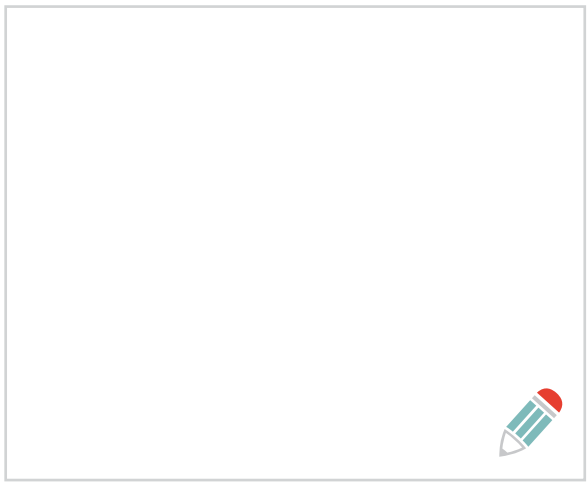
- How will I support students in retrieving this information?

- What types of feedback will assist students in determining metacognition?

- At what intervals will I space the retrieval?

- Am I able to interleave any of with previous learning and mix it up?

What is a **Power Ticket**? Draw an example or write down a description.



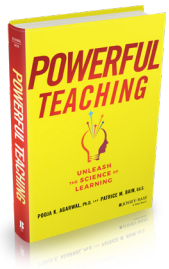
What are **mnemonics**?

- Come up with an example or a mnemonic you already use.

- Take a look at your curriculum. Can you find mnemonics and links that may help your students make connections that will aid retrieval?

What are the pros and cons of using Power Tools separately vs. combined?

Which Power Tools improve higher order thinking and transfer? Why?




# CHAPTER 7: KEEPING IT REAL: USE POWER TOOLS TO TACKLE CHALLENGES, NOT ADD TO THEM

What is one hesitation, challenge, or mountain between your current instruction and using Power Tools?

Work individually or in a group and identify one way Power Tools can address each of these challenges:

- Do I have to spend more time preparing for class if I use Power Tools?
- Do I have to spend more time grading if I use Power Tools?
- When I spend time using Power Tools, how can I cover the same amount of material?
- Will my students get lower grades if I use Power Tools?

- How can I adapt Power Tools for diverse students from a range of backgrounds and abilities?

- Do I have to spend money on Power Tools?

- There are so many Power Tools. Where do I start?

- There's a lot of stuff out there about "brain-based learning." Is that the same thing as Power Tools?

- When I use Power Tools, how can I find support beyond my school?

What is one more way to find support and accountability beyond your school or district as you apply Power Tools in your classroom?

What are the first five words that pop into your head when you think about the word "brain?"

1. _____ 2. _____ 3. _____

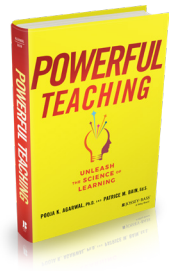
4. _____ 5. _____

| Neuromyths: True or False? | Circle One | |
|---|---|---|
| 1. We only use 10% of our brain. | T | F |
| 2. Children are less attentive after consuming sugary drinks and/or snacks. | T | F |
| 3. Short bursts of motor coordination exercises can improve integration of left and right hemisphere brain function. | T | F |
| 4. Students learn better when they receive information in their preferred learning style (e.g., auditory, visual, kinesthetic). | T | F |
| 5. The production of new connections in the brain can continue into old age. | T | F |
| 6. A common sign of dyslexia is seeing letters backwards. | T | F |
| 7. Some of us are "left-brained" and some are "right-brained" and this helps explains differences in how we learn. | T | F |
| 8. Listening to classical music increases children's reasoning ability. | T | F |
| 9. Academic achievement can be negatively impacted by skipping breakfast. | T | F |
| 10. Brain training games and crossword puzzles increase cognitive function and reasoning skills. | T | F |

**Answer Key:**

**1.**False **2.**False **3.**False **4.**False **5.**True **6.**False **7.**False **8.**False **9.**True **10.**False

See Chapter 7, note 9 (Macdonald et al., 2017) for more information.




# CHAPTER 8: FOSTERING A SUPPORTIVE ENVIRONMENT: USE POWER TOOLS TO REDUCE ANXIETY AND STRENGTHEN COMMUNITY

What's the difference between retrieval practice, formative assessment, and summative assessment?

In this chapter, we gave three ways student anxiety increases and three ways to decrease it. What are they?

| 1. | 2. | 3. |
|---|---|---|
| | | |

Do you remember what you ate for breakfast yesterday? Do you remember the name of the fourth president of the United States? How do these two questions make you feel? Do they feel fun or stressful? Why?

In this chapter, we presented our no-stakes Retrieval Warm Up strategy. What are one or two more Warm Ups you can create and share with your students?

First, what are the four Power Tools?

**1.** _____ **2.** _____

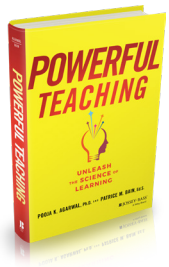
**3.** _____ **4.** _____

Second, which Power Tools are incorporated during Noun-Name Tag?

Contrary to our intuition, Power Tools – especially retrieval practice – have been proven to decrease student anxiety. Why?

Which of these strategies could you use frequently to create Retrieval Routines?

**A.** Brain Dumps **B.** Two Things **C.** Mini-Quizzes **D.** Dice Game




# CHAPTER 9: SPARK CONVERSATIONS WITH STUDENTS ABOUT THE SCIENCE OF LEARNING

In this chapter, we gave six steps for empowering students through conversations about the science of learning. What are the six steps?

| 1. | 2. | 3. |
|---|---|---|
| | | |
| **4.** | **5.** | **6.** |
| | | |

Second, we gave three ways to explain to students why Power Tools work. What are they?

1. _____
2. _____
3. _____

Third, we gave three ways to help students harness Power Tools outside the classroom. What are they?

1. _____
2. _____
3. _____

Contrary to our intuition, Power Tools - especially retrieval practice-have been proven to decrease student anxiety. Why?

What is an alternative name you can think of to describe spacing?

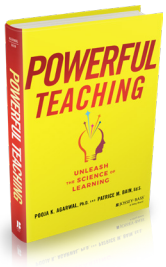
Do you remember who ruled Sumer?

- How long did he rule?
- Did you experience a desirable difficulty answering this question?
- Why or why not?

Have you been highlighting this book as you go along, or do you typically highlight when you read? Why?

What is one way you can help students form a concrete plan for using Power Tools at home?

How could you give an "elevator speech" about Power Tools to your students? Take a moment and outline it or write it out. (Bonus: Develop an activity where your students create a Power Tool elevator speech they can share with their parents or friends!)




# CHAPTER 10: SPARK CONVERSATIONS WITH PARENTS ABOUT THE SCIENCE OF LEARNING

What are three things you could say to parents about Power Tools, given the little time you have with parents?

1.

2.

3.

We presented five research snapshots you can share with parents. Describe at least one of the research studies in your own words.

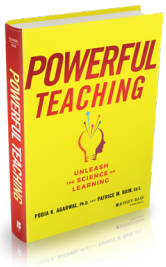
How did parents react to Patrice's nationwide survey?

› Were there any specific results that were surprising to you?

How can you set up your parent-teacher conferences, so the conversation centers around research-based learning strategies and not grades?

What is one reason you could share with parents about why Power Tools improve learning?

Research studies we present often include a restudy condition compared to a retrieval practice condition. Name one reason why restudy conditions are important in scientific research. If that was easy, challenge yourself: name two reasons!




# CHAPTER 11: POWERFUL PROFESSIONAL DEVELOPMENT FOR TEACHERS AND LEADERS

Take a moment and answer these two questions (you can only pick one answer per question!):

**11.** In the past two years, what is the main way you learn about education trends or new ideas that might be worth pursuing in your classroom?

**A.** Social media

**B.** Teacher-focused websites

**C.** News websites

**D.** Research Journals

**E.** Professional development and conferences

**F.** Colleagues and word of mouth

**12.** Which of the following is most important when it comes to deciding whether you will try a new idea in your classroom?

**A.** It aligns with my instructional approach

**B.** It's endorsed by my colleagues

**C.** It's endorsed by my school and district leadership

**D.** It's evidence-based

**E.** It's aligned with standards

› What do you predict was the most frequently selected choice for each question? Check Chapter 11 for the answers.

› Are your responses the same as your predictions for other teachers? Why or why not?

Imagine yourself as the teacher in this (slightly edited) proverb: Give an educator a fish and you feed them for a day. Teach an educator to fish and you feed them for a lifetime. Why is "teaching an educator to fish" important to you?

Imagine yourself as a leader, step by step: How will you support your colleagues as they make a small change from an old umbrella to a new umbrella? And what about a bigger change from an old house to a new house?

Imagine what you want your classroom, school, or district to look like when it's empowered with the science of learning. How might the culture look same or different?

Look around your classroom (or picture it in your mind).

- What does successful learning look like?
- What does unsuccessful learning look like?

Open up your plan book (or spreadsheet, or wherever you keep your lesson plans from the year) and just start browsing.

- How often do you space things out?
- How often do you use retrieval as a learning strategy, and not just as a formative or summative strategy?

Take a look at your student roster.

- Which of your students have high grades?

- Do they also have a keen sense of metacognition and awareness of their own learning?

- Which students struggle with retaining information?

Rose: What is one success you've had with Power Tools?

Thorn: What is one challenge you've had with Power Tools?

Bud: What is one new idea, strategy, etc. you've had with Power Tools?

Of the four Power Tools, what is one strategy you can implement tomorrow? What is one strategy you can implement next semester?

What are two challenges you may encounter in implementing Power Tools?

1.

2.

What are two unanswered questions you have?

1.

2.

Where are we now? What are we currently doing in this area?

- What are our most pressing needs in this area?

- Which one of the Power Tools would most likely help us meet our needs?

Where do we go from here? What steps can we take to implement this Power Tool?

- Who needs to be involved?

- What roadblocks might we encounter?

- How can we work around those roadblocks?

When it comes to the four Power Tools, what would you like yourself, your school, and your district to start, stop, and keep?

| Start | Stop | Keep |
|---|---|---|
| | | |

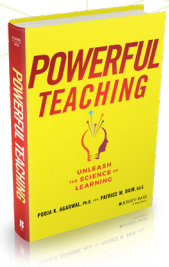
What are Two Things you've learned about each Power Tool?

1. _____

2. _____

What does powerful professional development look like to you? How does the science of learning fit in? _____

What is one additional way you can help a tutor, pre-service teacher, or fellow teacher become engaged in powerful teaching?



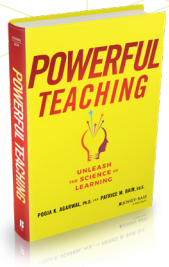

# CHAPTER 13: CONCLUSION: UNLEASH THE SCIENCE OF LEARNING

First, time yourself for five minutes and write a Brain Dump for everything you remember about this book!

Second, here are our learning objectives from the book introduction. Which objectives were met?

- Develop a deep understanding of powerful teaching strategies based on the science of learning, whether you are a past, present, or future educator
- Go behind the scenes and explore key findings from cognitive science research
- Gain insight into how scientifically-based strategies are effectively implemented in a variety of academic settings without additional preparation, classroom, or grading time
- Think critically about your current teaching practices and classroom environment from a research-based perspective
- Develop tools to share the science of learning with students and parents, ensuring success inside and outside the classroom
- Identify next steps to transform teaching and unleash the science of learning in your classroom

› Can you add one more thing you learned or achieved?




# RETRIEVAL WARM UPS AND EVERYDAY EXAMPLES

Here are our no-stakes Retrieval Warm Ups everyday examples of retrieval practice (i.e., trivia facts!). Just like for students (more details in Chapter 8 on building a supportive environment), these would be great to use at the beginning of professional development to get educators thinking about retrieval. You might have fun answering all of these right now, too!

- What did you have for breakfast yesterday?
- At what age did King Tut become a pharaoh?
- What was your favorite vacation?
- How many bones are there in the adult human body?
- In which U.S. city is the Baseball Hall of Fame located?
- In what year was the lawn mower invented?
- What is the name of the fourth president of the United States?
- What is your least favorite ice cream flavor?
- If you could travel anywhere in the world (where you haven't been before), where would you go?
- Would you rather visit the world 100 years in the past or 100 years in the future? Why?
- If you had to wear a hat every day for the rest of your life, what type of hat would it be?

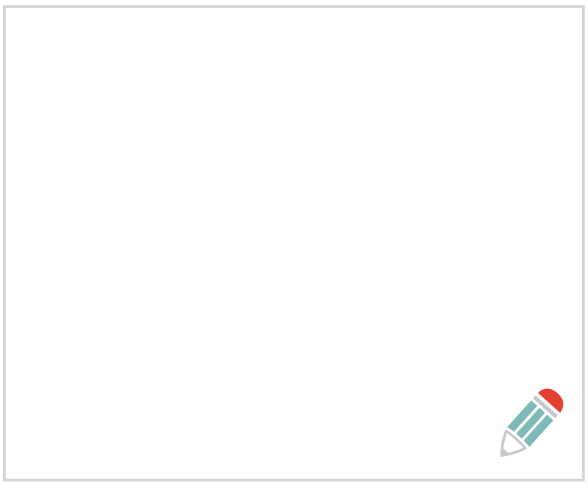
- Would you rather own a sailboat or a hot air balloon? Why?

- What is your favorite word? Why is it your favorite?

- If you could eat only one food or dish for the rest of your life, what would it be?

- What type of animal would be a really good musician? Why?

- What is your favorite emoji? Describe it or draw it!



- If you could switch places with anyone in your family for a day, who would it be?

- What is your proudest accomplishment this semester (or year)?

- Younger students: What is your favorite thing to eat for breakfast?

- Older students: What was your favorite thing to eat for breakfast as a kid?

- Would you rather lock yourself out of your car or out of your house?

- Have you ever gone skydiving? Why or why not?

- If you could choose only one thing from this class or professional development you want to remember in 10 years, what would it be and why?

